

1. Solution Description: Chain IIoT

Chain IIoT is a lightweight, all-in-one **cloud-edge automation platform** positioned as a practical and significantly lower-cost alternative to classical industrial automation stacks (PLC + industrial HMI + protocol gateway).

Core capabilities

- **All-in-one edge controller**

Integrates control logic, hybrid field networking, local operator interface and gateway functions into a single device. Because the platform is **web-native**, the edge controller itself can run on **commodity hardware**.

- **Hybrid field connectivity on the edge**

Native simultaneous support for **Modbus RTU (RS-485 multi-drop)** and **transparent LoRa**. The edge controller orchestrates and controls these local field networks with full real-time autonomy.

- **Cellular IoT (LTE-M / NB-IoT) support**

Cellular IoT devices are handled **directly by the cloud** and sit at the **same architectural level as edge controllers**. They interact with edge controllers (and with each other) through **MQTT messaging**. This enables seamless coordination between local edge-controlled zones (Modbus/LoRa) and wide-area or remote Cellular IoT nodes without forcing everything through a single gateway hierarchy.

- **Local autonomy first**

Real-time control loops, sequencing, interlocks and safety logic execute on the edge and continue during cloud or network outages.

- **Fully web-native architecture**

The entire stack — edge runtime, engineering/configuration, visualization and cloud interaction — is web-native (browser + app-script + on-device ML). Consequently:

- The edge controller can run on commodity hardware.
- Local and remote interfaces use ordinary browsers.
- Large dashboards use ordinary **Android / smart TVs** (or any browser-capable screen) in kiosk mode, served from the cloud.
- Most related devices can be commodity devices rather than expensive industrial hardware.
- No proprietary PLC programming environments or runtime licenses are required.

- **Cloud-based business processing & business-driven edge automation**

The cloud performs higher-level **business processing** (multi-site optimization, energy/tariff logic, service-level policies, scheduling, analytics, outcome tracking) and actively **drives both edge controllers and Cellular IoT devices** with business rules, setpoints and strategies. Real-time execution stays local where needed; business intent comes from the cloud. This enables true Automation-as-a-Service and outcome-based models.

- **Cloud-managed facility definitions**

Complete predefined facility packages (network topologies, scripting rules, ML models, setpoints, visualization layouts) — now covering combinations of edge (Modbus/LoRa) and Cellular IoT nodes — are stored and versioned in the cloud. During deployment or later in production, operators can simply **pick and switch** the appropriate definition.

- **Clean control / visualization split**

Edge controllers own local real-time control and include local touch interfaces (plus digital-twin style views). Large dashboards are cloud-served display surfaces using commodity screens.

Positioning

Chain IIoT targets the broad mid-market where classical industrial stacks are economically or operationally excessive. By remaining fully web-native and supporting both local hybrid field networks and cloud-direct Cellular IoT at the same architectural level, it delivers reliable control, wide-area reach and business-driven automation at substantially lower cost and complexity.

2. Comparison vs Classical Industrial Stacks

(Siemens, Schneider Electric, Rockwell Automation and equivalents)

Dimension	Classical Industrial Stack	Chain IIoT
Core hardware	Industrial PLC(s) + industrial HMI + gateway(s)	All-in-one edge controller on commodity hardware + cloud-direct Cellular IoT devices
Field / wide-area networks	PROFINET / EtherNet/IP / Modbus / BACnet; cellular usually via additional gateways or SCADA layers	Edge: native Modbus RTU + transparent LoRa ; Cloud-direct: LTE-M / NB-IoT at same level as edge; MQTT for inter-device messaging
Control logic	Vendor PLC languages + specialized IDEs	Fully web-native (app-script + edge ML)

Facility / project reuse	Templates possible; most sites still need significant engineering	Cloud library of complete definitions (edge + Cellular IoT) — pick & switch at deploy or runtime
Local autonomy	Strong (PLC continues offline)	Strong (edge continues offline); Cellular IoT coordinated via cloud + MQTT
Cloud role	Mainly telemetry, remote access, optional SCADA/analytics	Cloud-based business processing that actively drives both edge controllers and Cellular IoT devices
Operator / large display	Expensive industrial HMI panels + dedicated industrial monitors	Local touch on edge + ordinary browsers / commodity Android / smart TVs (cloud-served dashboards)
Device cost philosophy	Industrial-grade devices throughout	Most devices can be commodity (including the edge controller itself)
Cost profile	High CapEx + specialized engineering + licenses	Significantly lower hardware & integration cost
Deployment speed	Days to months	Faster, especially with pre-defined packages that mix edge and Cellular IoT
Skill barrier	Specialized OT / PLC engineers	Lower — web-configurable + selectable packages
Service model fit	Primarily product / project sales	Designed for Automation-as-a-Service and business-outcome models
Best scale fit	Medium to very large / high-criticality processes	Small to mid-scale, multi-site, cost-sensitive, retrofit, and mixed local + wide-area deployments

Key differentiators of Chain IIoT

1. Fully web-native architecture allows the edge controller and most related devices to be commodity hardware.
2. Cellular IoT (LTE-M / NB-IoT) devices are first-class peers of edge controllers, handled directly by the cloud and interacting via MQTT — not subordinated through extra industrial gateways.
3. Cloud-managed facility definitions enable rapid, low-engineering replication of proven mixed (edge + cellular) configurations.
4. The cloud performs real **business processing** and drives both edge and Cellular IoT nodes with business intent.

3. Domains They Apply Respectively

Classical stacks remain preferable when:

- Process criticality or regulations demand certified, highly deterministic PLC platforms with extensive functional safety (SIL) options
- Installations are large, complex and highly customized
- There is already deep investment in a specific vendor ecosystem
- Hard real-time motion control or complex continuous process control is required
- Customer specifications explicitly mandate industrial PLC platforms

Typical domains: Heavy process industries, large discrete manufacturing, major critical infrastructure, large hospitals, large campus BMS.

Chain IIoT is preferable when:

- Classical stacks are economically or operationally excessive for the scale of the site
- Rapid replication of similar facilities is needed
- Hybrid local (Modbus + LoRa) and wide-area (Cellular IoT) connectivity is valuable
- Multi-site operators want central **business processing** and policy management with local autonomy
- Automation-as-a-Service or outcome-based commercial models are desired
- Visualization and even the controller itself can use commodity devices
- Engineering resources are limited and configurable packages are preferred
- Sites mix dense local control zones with remote or sparsely distributed Cellular IoT nodes

The two approaches are **complementary**.

4. Addressable Markets for Chain IIoT

(with typical problems and why Chain IIoT fits, including Cellular IoT value)

Primary Markets

1. Controlled Environment Agriculture (CEA)

Greenhouses, high/low tunnels, vertical farms

- **Typical problems:** Full industrial climate computers are too expensive for many mid-sized growers; multi-site recipe/climate consistency is hard; wireless expansion and remote sites add cost.

- **Why Chain IIoT fits:** Commodity edge control + Modbus/LoRa for dense greenhouse zones; Cellular IoT for remote sensors, weather stations or distant tunnels; cloud facility packages and business processing for multi-site optimization and AaaS.

2. Smart Care / Elderly Care / Senior Living

Assisted living, nursing homes, memory care

- **Typical problems:** Limited budgets and OT expertise; classical BMS overkill; need environmental control, simple sensing and status visibility; many sites already have Android TVs.
- **Why Chain IIoT fits:** Low-cost commodity edge for HVAC and local sensors; LoRa for in-building wireless; Cellular IoT for outdoor, campus-wide or roaming devices; cloud-served dashboards on existing TVs; business processing for multi-site care operators.

3. Smart Buildings (small-to-medium commercial)

Offices, retail, schools, clinics, smaller hotels

- **Typical problems:** Full BMS often over-specified and too expensive; multi-building portfolios lack affordable central policy and energy optimization.
- **Why Chain IIoT fits:** Lightweight multi-protocol edge on commodity hardware; Cellular IoT for meters, outdoor units or remote buildings; cloud business processing for portfolio energy/tariff optimization and service packages.

4. Smart Communities

Gated communities, apartment complexes, senior-oriented housing

- **Typical problems:** Shared systems need distributed control; classical stacks rarely justified; operators want central visibility and service packages.
- **Why Chain IIoT fits:** Distributed edge nodes for dense zones + Cellular IoT for community-wide or remote assets; cloud orchestration and facility packages; existing TVs as dashboards; business-driven shared-resource optimization.

5. Aquaculture

Fish/shrimp farms, recirculating systems

- **Typical problems:** Water quality, pumps, aeration need reliable control; industrial PLC solutions often too costly; multi-site and remote monitoring desired.
- **Why Chain IIoT fits:** Edge for local intensive systems; Cellular IoT for ponds, cages or remote sensors; cloud packages and business processing for multi-farm management and services.

6. Cold Chain & Food Storage

Cold rooms, refrigerated warehouses, produce storage

- **Typical problems:** Continuous monitoring and simple control critical; multi-site compliance required; classical SCADA expensive for mid-sized or distributed facilities.
- **Why Chain IIoT fits:** Commodity edge for local control; Cellular IoT for remote or mobile cold units; cloud business processing for compliance, alerts and multi-site reporting.

Secondary / Adjacent Markets

- **Open-field & broader agriculture** — Long-range sensing and pump control; Cellular IoT + LoRa excel for wide areas; cloud multi-farm business logic.
 - **Water & wastewater (small/mid-scale)** — Classical SCADA too expensive for smaller systems; Cellular IoT for remote stations + edge for local plants; managed services attractive.
 - **Energy edge** (solar + storage, microgrids, EV clusters) — Modbus-heavy equipment + Cellular IoT for distributed assets; cloud tariff optimization and aggregation.
 - **Hospitality & multi-site retail** — Multi-site consistency and cost pressure; edge + Cellular IoT + cloud business policies + commodity dashboards.
 - **Light industrial / small manufacturing** — Full industrial stacks excessive; affordable commodity edge + selective Cellular IoT.
 - **Education & campus facilities** — Multi-building, budget-constrained; distributed edge + Cellular IoT + central cloud management.
 - **Municipal / smart-city edge** — Fragmented, incremental deployments; Cellular IoT and LoRa + commodity edge nodes ideal for phased roll-outs.
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Summary

Chain IIoT delivers reliable local control (Modbus RTU + transparent LoRa on the edge), wide-area reach (cloud-direct Cellular IoT at the same architectural level, coordinated via MQTT), fully web-native commodity-hardware economics, cloud-managed facility definitions, and **cloud-based business processing that drives both edge and Cellular IoT devices**.

It is the practical choice for the broad mid-market of CEA, smart care, smart buildings, smart communities and related cost-sensitive, multi-site or mixed local/wide-area domains — where classical industrial stacks are economically or operationally out of reach.